The Subset-Vanishing Theorem: $B \equiv 0 \;\Rightarrow\; B_i \equiv 0$*

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Section 3 — The Subset-Vanishing Theorem

This section gives a one-paragraph proof of the implication

$$B(x,y) \equiv 0 \text{ on } \{\ell(x)=0\} \quad \Longrightarrow \quad B_i(x,y) \equiv 0 \text{ on } \{\ell(x)=0\} \text{ for every } i,$$

where B is partitioned by homogeneity in the x-variables and ℓ is any linear map of x alone. This is the result that lets one work zone-by-zone in the BCFW grading without losing information, and it is the workhorse of every uniqueness argument in the bootstrap. The reason for stating it as a stand-alone theorem with minimal hypotheses is that the standard reference [1, eqs. (15)–(18)] reaches the same conclusion through a chain of structural claims about $X^{(0)}$ vs. $X^{(\infty)}$ identities (the latter requiring his Appendix A); the proof below shows that this chain is unnecessary — a single rescaling argument suffices.

3.1 Statement and proof

Definition 1 (Homogeneity partition)**.** Let $B(x,y)$ be a rational function with $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$, and finitely many monomials. Group the monomials of B by total degree in x:

$$B(x,y) \;=\; \sum_{i\in\mathbb{Z}} B_i(x,y), \qquad B_i(\lambda x,\, y) \;=\; \lambda^i\, B_i(x,\, y) \text{ for all } \lambda \neq 0.$$

The sum is finite. We call $\{B_i\}$ the *x-homogeneity partition* of B.

Theorem 2 (Subset vanishing)**.** *Let $B(x,y)$ be a rational function with x-homogeneity partition $B = \sum_i B_i$, and let $\ell : \mathbb{R}^n \to \mathbb{R}^k$ be a linear map (so ℓ depends on x only). Then*

$$\Big[\,\ell(x)=0 \;\Rightarrow\; B(x,y)=0\,\Big] \qquad \Longleftrightarrow \qquad \Big[\,\ell(x)=0 \;\Rightarrow\; B_i(x,y)=0 \textit{ for every } i\,\Big].$$

Proof. "$\Leftarrow$" is trivial since $B = \sum_i B_i$.

For "$\Rightarrow$", fix any (x,y) with $\ell(x)=0$ and any nonzero $\lambda \in \mathbb{R}$. By linearity, $\ell(\lambda x) = \lambda\,\ell(x) = 0$, so $(\lambda x, y)$ also lies on the zero locus. The hypothesis gives $B(\lambda x, y) = 0$. Expanding,

$$0 \;=\; B(\lambda x, y) \;=\; \sum_i B_i(\lambda x, y) \;=\; \sum_i \lambda^i\, B_i(x,y).$$

The right-hand side is a Laurent polynomial in λ that vanishes at every nonzero λ. A Laurent polynomial vanishing on an infinite set has every coefficient equal to zero, so $B_i(x,y) = 0$ for every i. □

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Remark. The proof uses only that ℓ acts linearly on the homogeneity variables. Nothing is assumed about the structure of ℓ, the relations the x_i satisfy on the locus $\{\ell(x) = 0\}$, or the algebraic form of B. In particular, no claim is made or needed that B lies in the ideal generated by ℓ.

3.2 Connection to Rodina's eqs. (15)–(18)

In [1, "Subset zeros and UV scaling"], the same equivalence appears in the specific setting where B is a rational function of planar Mandelstams $\{X_{ij}\}$, the locus is a hidden k-zero $\{c_{ij} = 0\}$, and the homogeneity grading is the BCFW-shift scaling under $p_{k+1} \to p_{k+1} + zq$, $p_n \to p_n - zq$. With the dictionary

$$x \;=\; (z\text{-dependent } X_{ij}), \qquad y \;=\; (z\text{-independent } X_{ij}), \qquad \ell(x) \;=\; \text{the } c_{ij} \text{ relations restricted to } x,$$

Theorem 2 is precisely his eq. (14), restricted to "B satisfies the zero $\Leftrightarrow$ each B_i satisfies the zero."

Rodina's argument. The chain in [1, eqs. (15)–(18)] runs through the structural form of B on the zero locus. Eq. (15) asserts that $B(X_{ij}^{(0)}) = 0$ implies $B = \sum_{c_{ij}\in\text{zero}} c_{ij}\,\{\dots\}_{ij}$, on the grounds that "the only identities the $X_{ij}^{(0)}$ satisfy are given by $c_{ij} = 0$." BCFW-shifting and taking $z \to \infty$ (eq. 16) gives $B \to z^m B_m(X_{ij}^{(\infty)})$. Eq. (17) then claims that B_m also has a representation $\sum c_{ij}\{\dots\}_{ij}$ *with the same* c_{ij}, on the grounds that "$X_{ij}^{(\infty)}$ and $X_{ij}^{(0)}$ satisfy identical relations" — a non-obvious claim established by the basis-counting in his Appendix A. From eq. (17) the leading B_m peels off, the procedure iterates on $B - B_m$ (eq. 18), and one extracts each B_i in turn.

The argument is correct, but it has three moving parts: (1) the structural representation in eq. (15); (2) the identification of $X^{(\infty)}$ relations with $X^{(0)}$ relations (Appendix A); (3) the iterative peel-off (eqs. 17–18).

What the rescaling proof eliminates. Theorem 2 reaches the same conclusion using only one move: rescale $x \mapsto \lambda x$ inside the zero locus (legal because ℓ is linear), then read off Laurent coefficients in λ. In particular:

- No structural representation $B = \sum c_{ij}\{\dots\}_{ij}$ is invoked. The hypothesis is the bare equation $\ell(x) = 0 \Rightarrow B = 0$; we never write B as an explicit element of any ideal.
- No comparison of $X^{(0)}$ and $X^{(\infty)}$ relations is required. The argument stays inside one fixed locus throughout. *Appendix A of [1] is therefore not on the critical path for this equivalence.*
- No iteration is needed. All B_i vanish simultaneously from a single application of the Laurent argument.

Where Appendix A is still needed. [1, Appendix A] proves that $X_{ij}^{(\infty)}$ and $X_{ij}^{(0)}$ satisfy the same set of $c_{ij} = 0$ identities. This fact remains useful elsewhere in his program — specifically for the third equivalence in eq. (14), the statement that subsets exhibit *enhanced* UV scaling z^{i-1} rather than z^i. Theorem 2 does not reach the enhanced-scaling refinement; it gives only "B satisfies the zero $\Leftrightarrow$ each B_i does." The enhanced-scaling step compares the z^{m-1} coefficient on the limit locus to the c_{ij} ideal on the zero locus, and that comparison genuinely needs the Appendix A identification. We make no claim against that part of [1]; the point is to make the dependency explicit and isolated.

References

[1] L. Rodina, "Hidden zeros are equivalent to enhanced ultraviolet scaling and lead to unique amplitudes in $\mathrm{Tr}(\phi^3)$ theory," arXiv:2406.04234.